

concentration of Prime subscribers was large enough to justify operational costs.

However, it soon became an issue when people realised that predominantly non-white neighbourhoods were primarily excluded from this offering. Customers from these areas were not happy.

When this issue was highlighted in several forums and the media, Craig Berman, Amazon's vice president for global communications, asserted, "When it comes to same-day delivery, our goal is to serve as many people as we can." He continued, "Demographics play no role in it. Zero." From Amazon's standpoint, this seemed to be a logical approach from the cost and efficiency perspective, such that they would prioritise areas with most existing paying members over the others.

A solely data-driven approach that looked at numbers instead of people did reinforce long-entrenched inequality in access to their services. For people who were (inadvertently) excluded, it did not make much difference. The fact that racial demographics tend to correlate with location or zipcode did result in indirect discrimination.

Apparently, in scenarios where zipcodes are considered as one of the data points for any decision-making, there will always be some inherent bias. It is because zipcodes usually represent tightly-knit communities of one kind. Perhaps better sampling of input data could help to a certain extent in minimising this risk, but it would be far from being absolute zero.

I remember about ten years ago, Indian banks used this (zipcode-based) strategy to exclude applicants from credit card or personal loan applications. I do not know if they are continuing these practices still.

Uber's autonomous car kills a pedestrian

An Arizona pedestrian was killed by an automated vehicle owned by Uber, in 2018. A preliminary report released by National Transportation Safety Board (NTSB) in response to the incident stated that there was a human presence in the automated vehicle, but the human was not in control of the car when the collision occurred.



National Transportation Safety Board investigators examine Uber's self-driving vehicle that was involved in a fatal accident in Tempe, Arizona, in March 2018 (Credit: <https://cnet4.cbsiatic.com>)

An accident usually implies several things going wrong at once and, hence, it is difficult to investigate as well as ascertain final accountability and root cause. Various reasons contributed to this accident, including poor visibility of the pedestrian, inadequate safety systems of the autonomous car and lack of oversight by the human assistant.

The legal matter was eventually settled out of courts, and further details were not released. However, issues around liability present significant complexity due to multiple parties and actors involved. On this occasion, the vehicle was operated by Uber but was under the supervision of a human driver who was an Uber employee, and it was operated autonomously using components and systems designed by various other tech companies.

Responsibility attribution with an AI system poses a significant dilemma and, unfortunately, there are no set standard guidelines to make it more transparent.

On occasions like these, where human lives are directly affected, clarity is undoubtedly needed along with a universal framework to help in a standardised approach to deal with such matters.

Although there was no final and clear legal verdict as such, the fact

that ambiguity exists is a significant concern. When there are several actors at play, human as well as non-human, direct as well as indirect, ascertaining accountability, root cause and liability will be the key for clarity.

DoS attack on smart home by a light bulb

In 2009, Raul Rojas, a computer science professor at Free University of Berlin, built one of Germany's first smart homes. Everything in his house was connected to the Internet so that lights, music, television, heating and cooling could all be turned on and off from afar. Even the stove, oven and microwave could be turned off with Rojas's computer.

Now the challenge with his setup was an ubiquitous one—there were several manufacturers and protocols at play. To help address this, Rojas designed the whole system to connect all smart devices to one hub. The hub was the central coordinator of all communication between devices as well as through the Internet.

A few years after the installation, during 2013, the smart home gave up and stopped responding to Rojas' commands. In computer parlance, the system hung-up or froze. When Rojas investigated, it turned out that there was just one culprit which caused this problem, a connected lightbulb!